Chapter 1: Semiconductor Physics and PN Junctions.

A P-N junction is formed when a p-type semiconductor is joined to an n-type semiconductor.
At the junction, free electrons and holes recombine, creating a depletion region devoid of mobile carriers.
For Silicon, the built-in barrier potential is approximately 0.7 Volts.
In forward bias, current increases exponentially with applied voltage.

Chapter 2: Bipolar Junction Transistors.

A BJT consists of three doped regions: Emitter, Base, and Collector.